

MAT 426, Class 9 Homework: Orthogonal Matrices & Decomposition
(Due Tuesday, March 10, 2026)

1. Determine whether the given matrix is orthogonal. If it is, find its inverse.

(a)
$$\begin{bmatrix} \frac{1}{3} & \frac{1}{2} & \frac{1}{5} \\ \frac{1}{3} & -\frac{1}{2} & \frac{1}{5} \\ -\frac{1}{3} & 0 & \frac{2}{5} \end{bmatrix}$$

(b)
$$\begin{bmatrix} \cos(\theta) \sin(\theta) & -\cos(\theta) & -\sin^2(\theta) \\ \cos^2(\theta) & \sin(\theta) & -\cos(\theta) \sin(\theta) \\ \sin(\theta) & 0 & \cos(\theta) \end{bmatrix}$$

(c)
$$\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

2. Let U be an $n \times n$ orthogonal matrix. Show that the rows of U form an orthonormal basis of $\mathbb{R}^n$.
3. Let U and V be $n \times n$ orthogonal matrices. Prove that UV is an orthogonal matrix.
4. Let U be an orthogonal matrix.
- (a) Show that U^{-1} is an orthogonal matrix.
- (b) Show that $\det(U) = \pm 1$.
- (c) If λ is an eigenvalue of U , then $|\lambda| = 1$.
5. Prove that if an upper triangular matrix is orthogonal, then it must be a diagonal matrix.